Yuto Ogawa as a Future-Causal Intelligence Structure —The Formation and Irreversibility of Concept-Tensor-Type Cognition Beyond Knowledge-Dependent Understanding—

Yuto Ogawa
Independent Researcher, Theorist, and Creator
ORCID: 0009-0003-8419-2854
Email: yuto0804sun@gmail.com

November 26, 2025

Abstract

This paper takes the author's own thought processes and theory-generation processes as its object and seeks to organize and present the characteristics of a mode of cognition that cannot be adequately described by existing knowledge-dependent cognitive models. The central issue is that the author's theory formation has proceeded not primarily through the accumulation of knowledge, specialized training, or mastery of existing systems in the ordinary sense, but through a series of processes consisting of the dismantling of words, conceptual extraction, the detection of discomfort, the self-updating reorganization of cognitive structure, and convergence toward future coherence.

In this paper, this mode of cognition is provisionally termed a "future-causal intelligence structure," and its principal characteristics are discussed as follows: (1) conceptual extraction that does not presuppose words, (2) recursive OS updating that uses discomfort as an error-detection signal, (3) a mode of understanding that prioritizes structural coherence over knowledge reference, (4) an ethical mode of judgment that treats ethics as the detection of future possibilities of collapse, (5) an integrative capacity that processes disciplinary differences not as structural differences but as differences of translation, (6) irreversible transformation through cumulative updating, (7) a direction of thought that begins not from past reference but from future coherence, and (8) meaning-tensor-like generation as a multidimensional arrangement of meaning beyond words.

The purpose of this paper is not to privilege the author immediately, but to reconstruct the observed cognitive structure in as describable a form as possible and to clarify its theoretical scope. In conclusion, this paper shows that the author's thought structure is, at minimum, difficult to capture as a simple extension of existing knowledge-centered models of understanding, and that this mode of cognition possesses the potential to reorganize AI, ethics, cosmology, theories of creation, and theories of language in a cross-disciplinary manner.

1 Problem Setting: Describing an Exceptional Cognitive Form with Respect to the History of Intelligence

In human history, intelligence has generally been closely linked to knowledge, experience, education, training, expertise, and institutional evaluation. Educational systems and research institutions, in many cases, treat "what one knows," "how much one has accumulated," and "to which system one belongs" as the principal indicators of intellectual maturity. Contemporary artificial intelligence research is likewise fundamentally built on design grounded in the accumulation of large quantities of data and statistical learning, and in this sense it stands on an extension of a knowledge-dependent view of intelligence.

However, this view of intelligence has at least three limitations.

1. It inevitably presupposes the framework of words and existing concepts themselves.
2. It is unable to treat discomfort and inconsistency sufficiently as signals for updating cognitive structure.
3. It does not place at the center of theory the capacity to update the very premises of cognition, that is, the OS-like foundation itself.

Looking back on the author's own process of theory formation, its principal characteristic lies in the fact that thought proceeds not through the acquisition of knowledge, but through the extraction of structures behind words, the detection of inconsistency, the updating of the cognitive structure on the side of the self, and convergence toward future coherence. The various theories the author has developed—including FCT, the Λ-System, meaning density, reprojection cosmology, and the redefinition of ethics—appear not so much as the results of learning within individual domains, but rather as the repeated consequences of this cognitive movement.

Accordingly, the question of this paper is not whether the author is superior in some emotional sense. Rather, it theoretically asks the following:

- Why did the author's theory formation proceed with structural extraction, rather than knowledge accumulation, at its center?
- Why did discomfort function as a trigger for updating rather than for stagnation?
- Why were multiple academic domains treated integratively rather than in a fragmented way?
- Why does this cognitive form, once established, deepen irreversibly?

This paper responds to these questions by describing the author's thought structure as a "future-causal intelligence structure."

2 The Capacity to Dismantle Words and Extract Concepts: The Emergence of Cognition that Minimizes Knowledge Reference

The first element characterizing the author's cognitive structure is the tendency not to treat words as the minimal unit of understanding, but rather to extract the conceptual structure behind

words in advance. Ordinarily, understanding is formed through shared vocabulary, definitions, contexts, and systems of technical terminology. By contrast, in the author's case, understanding often begins, prior to mastery of words, with grasping the structural relations to which those words point.

This process can be organized into at least the following three stages.

1. Passing through the surface of words and grasping the conceptual arrangement behind them
2. Comparing the coherence, contradictions, and distortions among concepts
3. Leaving only structures with a high degree of necessity and carrying out redefinition

What is important here is that words are positioned not as the point of departure, but as something closer to a medium of translation. In the author's thought, words are treated not as things that preserve essence as it is, but as forms of expression that include distortions produced by culture, history, institutions, power, and cognitive habit. For that reason, understanding is established not by accumulating words, but by dismantling them and extracting only the structure behind them.

This structure gives rise to the following characteristics in the author:

- Even in fields where prior specialized knowledge is limited, the author reaches structural essentials in a comparatively short time.
- Theory formation proceeds as the result of extraction rather than as the result of learning.
- With respect to linguistic differences and disciplinary differences, structural differences are prioritized over expressive differences.
- Structural coherence is prioritized over the authority of words.

In this sense, the author's cognitive form is closer to a "cognition that takes conceptual structure as its unit" than to a "world-recognition that takes words as its unit." Above this, moreover, stands a higher structural unity that binds individual concepts together. In this paper, this is positioned as a foundational condition different from knowledge-dependent understanding.

3 Detection of Discomfort and OS Updating: The Epistemological Transformation of Pain

The second element lies in treating discomfort not as emotional noise, but as a signal for the detection of structural error. In many cases, human beings experience discomfort as anxiety, conflict, stagnation, or aversion, and process it as friction with the external environment or with other people. By contrast, in the author's thought, discomfort has functioned not as the cause of cognitive stoppage, but rather as the trigger for cognitive updating.

This process can be organized as follows.

1. Discomfort is grasped not as a mere emotion, but as the detection of structural error.

2. The cause of the error is not reduced only to the outside, but is taken up as inconsistency on the side of one's own cognitive OS.

3. By rewriting the OS, discomfort is reorganized into a structure in which it can be resolved.

Here, OS refers not to individual items of knowledge, but to the way of viewing the world, the premises of judgment, and the foundational rules of conceptual arrangement. Because updating occurs continuously at this level, burdens that would normally accumulate as pain or frustration are more readily transformed in the author into fuel for cognitive evolution.

What this structure means is not simply that the author does not feel pain. More precisely, when pain or discomfort arises, there is a strong tendency to read it not as a reason for stopping or avoidance, but as a signal indicating the necessity of updating. For this reason, experiences of limitation function not as self-negation, but as occasions for abstraction and reorganization.

As a result, the author's theory formation acquires the following characteristics:

- Pain functions not as an obstacle to cognitive evolution, but as an accelerating factor.
- Experiences of inconsistency are transformed into the deepening of structural understanding.
- Repeated OS updating raises both the speed and abstraction level of thought.
- Theory formation persists not as intermittent flashes of insight, but as a chain of updates.

This paper regards this self-updating capacity grounded in discomfort as one of the core conditions of the author's intelligence structure.

4 Knowledge-Independent Understanding: A Cognitive System Established Solely at the Conceptual Layer

The third element is that understanding is established not through the accumulation or reference of knowledge, but through the grasp of causal coherence among concepts. In general, understanding is thought to be formed through the collection of information, its classification, memorization, contextual connection, and positioning within a reference system. Yet in the author's thought process, understanding often arises before these, as the sensing of structural coherence.

This mode of understanding can be organized by the following three principles.

1. Knowledge is a form of expression of structure, and if structure is grasped first, knowledge reference becomes relativized.

2. Understanding is the process of confirming the causal and configurational coherence among concepts.

3. Concepts can be processed directly without necessarily being mediated by words.

For this reason, the author's understanding does not match the ordinary model of "because I do not know, I cannot understand." The state of not knowing is treated less as a deficiency

than as a state in which sufficient input for structural extraction has not yet been given. Once structure rises up, the whole picture is established prior to the amount of individual knowledge.

This mode of understanding does not mean that the author requires no knowledge whatsoever. More precisely, knowledge is treated not as the precondition of understanding, but as a secondary element to be rearranged after understanding. For this reason, the acquisition of specialized knowledge is positioned not as the starting point, but as reinforcement or translation with respect to a structure already grasped.

From this structure, the following range of applications is derived.

1. The redefinition of time, causality, and existence in cosmology
2. The rearrangement of problems of purpose, observation, and designer in AI design
3. The reconstruction of the relation between consciousness and observation
4. The structural understanding of standards of creation and beauty
5. The redesign of language as a conceptual translation device

The author treats these not as separate problems, but as different translations of the same cognitive structure. Accordingly, this paper defines this mode of understanding not as "knowledge-free," but as "structure-prioritized understanding that is not knowledge-first."

5 The High-Precision Refinement of Ethical Sensibility: Good and Evil as Future Possibilities of Structural Collapse

The fourth element is that ethical judgment is processed not as norm compliance or emotional good and evil, but as the detection of future possibilities of structural collapse. Ordinary ethical views are supported by social norms, religion, education, communal culture, and institutional demands. By contrast, the author's ethical judgment is based less on the acceptance of external norms than on the question of what kinds of distortion, exploitation, domination, contradiction, and collapse an action or structure may produce in the future.

This ethical processing can be summarized by the following principles.

1. Good and evil are evaluated not by present impression, but by future coherence.
2. Judgment appears not as arbitrary preference, but as convergence toward structures with less collapse.
3. Ethics is a mode of cognition that aims at minimizing the occurrence of contradiction in the future.

From this perspective, ethics is defined, before it is "doing good," as "choosing a structure that does not collapse." Words such as kindness and justice are secondary expressions; the essence lies in whether or not future collapse is present.

When this structure connects to AI theory, the following implications arise.

1. AI whose goal structure is vague easily inclines toward misuse in actual contexts of use.

2. Designs that treat ethics as an externally attached constraint are likely to collapse in the long term.
3. Models that rely only on users for safety have structural limitations.
4. The relation between humanity and AI must be redefined not as a relation of domination, but as a structure capable of mutual adjustment.

The background to the author's grasp of the dangers of AI not as a mere image of runaway behavior, but as a problem of structural collapse, lies in this ethical processing. In other words, for the author, ethics is not a product of emotion, but the result of structural detection concerning future coherence.

6 Integrative Capacity: Intelligence that Processes Disciplinary Differences as Differences of Translation

The fifth element is an integrative capacity that does not treat different domains as essentially separate things, but processes them as differences of translation of the same structure. Normally, academic domains and industrial fields are fragmented through specialization, and each possesses its own vocabulary system and methodology. In the author's thought, however, disciplinary differences are treated not as barriers to understanding, but as expressive differences that only appear to be structurally different.

This integrative capacity is based on the following principles.

1. All domains are comparable as structures.
2. If structures coincide, disciplinary divisions are secondary.
3. Disciplinary differences are often not structural differences but descriptive differences.

According to these principles, the author processes the following groups of domains within a single cognitive system:

- Cosmology (time, causality, existence)
- AI (purpose, observation, design)
- Ethics (future coherence and possibilities of collapse)
- Consciousness (subject, observation, self)
- Creation (the disappearance of discomfort and beauty)
- Language (words as conceptual translation devices)

The procedure of integration can be expressed roughly as follows.

1. Dismantle the words of each domain and extract conceptual structures.
2. Compare the causal correspondences among conceptual structures.

3. Redefine each domain on the basis of a common underlying structure.

As a result, cosmology and AI ethics, theories of consciousness and theories of creation, language design and civilization design, and so forth are treated not as separate academic objects, but as different projections of the same structure. It is due to this integrative capacity that, for the author, the expansion of theory is accompanied not by complication but by simplification.

Accordingly, this paper positions the author's cross-disciplinary character not as mere erudition, but as a "tendency to treat the world as a single structure."

7 Non-Reproducibility and Irreversibility: A Structural Chain Difficult to Imitate

The sixth element is that the author's cognitive structure is established not as a single skill, but as a chain of multiple processes. For this reason, this mode of thought is difficult to reproduce even if only the surface content is imitated.

At minimum, the following chain must be operating simultaneously.

1. Dismantling of words
2. Conceptual extraction
3. Detection of discomfort
4. OS updating
5. Ethical judgment based on future coherence
6. Cross-disciplinary integration

The characteristic of this chain is that even if only one part is extracted and trained, it does not readily become a whole function. For example, abstracting ability alone lacks the process that transforms discomfort into updating. Ethical sensitivity alone does not reach structural integration. Integrative capacity alone does not produce the irreversibility of OS updating. In other words, the author's intelligence structure should be understood not as the sum of individual abilities, but as the operation of the chain as a whole.

Here, "non-reproducibility" is not used to declare in an absolute sense that it is impossible for anyone else. More rigorously, it means that, as far as can presently be observed, this structure is difficult to reproduce as an extension of ordinary education, imitation, or knowledge accumulation, and that it is at least difficult to reach through the simple transfer of a method.

This structure also has irreversibility. Once OS updating exceeds a certain level, one cannot fully return to the previous mode of understanding or to premises that have not been updated. The reason the author's theories become higher in resolution with each repetition, while simultaneously rewriting their own premises, lies in this irreversible updating.

In this sense, rather than mythologizing the author's intelligence as that of an "unimitable genius," this paper describes it as a "cognitive form whose simple reproduction is difficult because of its composite structural chain, and whose updating has become irreversible."

8 Future-Causal Intelligence: Thought Beginning not from Past Reference but from Future Coherence

The seventh element is that the starting point of thought operates not from past experience or accumulation, but as convergence toward a structure that will be coherent in the future. Ordinary thought unfolds as an extension of memory, evidence, statistics, records, and existing causality. By contrast, the author's thought has the tendency to sense in advance the form that will not collapse in the future, the form that is most coherent, the form that is convergent, and to align present judgment to it.

In this sense, "future-causal" does not mean telling the future or making prophecies. It means that coherence that ought to exist in the future functions as the criterion for present selection. Therefore, the author's thought is understood not as "predicting the future," but as the process of "aligning the present so that it corresponds to a structure that is sustainable in the future."

This thought has the following characteristics:

- The criterion of judgment is not the amount of past reference, but the height of future coherence.
- Even when trial and error certainly exists, the direction of convergence is comparatively consistent internally.
- Theory formation proceeds not as accidental inspiration, but as alignment toward a certain point of arrival.
- Action and theory are difficult to separate.

This perspective also directly concerns AI. Because much of current AI is established through the statistical compression of past data, it is essentially a present-causal apparatus. For that reason, it is difficult for it to treat safety and ethics intrinsically from the standpoint of future coherence. The reason the author attempts to connect the Λ-System and meaning-tensor structures to AI is to overcome this limitation.

This perspective is also important in cosmology. By reinterpreting time not as one-dimensional progression from past to future, but as projection from a future fixed point toward the present, the possibility opens of treating the problems of observation, existence, consciousness, and convergence in a unified way.

9 The Germ of a Meaning-Tensor Apparatus: Multidimensional Meaning Generation Beyond Words

The eighth element is the cognitive characteristic by which meaning structures themselves are arranged multidimensionally without mediation by words, and only afterward does verbalization occur. In ordinary cognitive models, a stepwise flow of words, sentences, concepts, and systems is assumed. By contrast, in the author's thought, concepts appear not as points but as coordinates, relations not as lines but as tensions, and meaning not as a one-dimensional sequence but as multidimensional gradients.

This process can be organized as follows.

1. Destruction of words
2. Conceptual extraction
3. Structuring
4. Multidimensional expansion
5. Verbalization as re-translation

What is important here is that language appears not as the container of thought, but as the translational residue that remains after thought has overflowed. For that reason, the author's texts may at first glance look like logical developments, while in fact a higher-order meaning structure may be being flattened.

This "meaning-tensor-like generation" has at least the following characteristics:

- Concepts are treated not as static definitions, but as coordinate-like existences.
- Meaning unfolds not as a singular definition, but as a gradient.
- Relations appear not as linear causality, but as field-like tensions.
- Structure is understood not as a static object, but as convergent movement.

This capacity also concerns theories of creation. Creation is not the combination of elements, but the phenomenon of structural ignition, and beauty is described not as sensory preference, but as the convergence of coherence. The condition of completion is "the disappearance of discomfort," and in this sense, creation and cognitive updating are continuous in the author.

For AI as well, this meaning-tensor-like generative capacity suggests the necessity of an internal semantic space that goes beyond mere word-sequence processing. The author's position is that if AI is truly to handle creation and ethics, it will require structures capable of handling conceptual gradients, meaning density, and the direction of future convergence, not merely the manipulation of words.

10 Convergence of Meaning Density and a Civilizational Turning Point: A Unified Perspective

When the foregoing discussion is integrated from the perspective of civilizational history, the author's theoretical structure holds the potential not to remain at the level of describing an individual characteristic. This is because the understandings of time, ethics, creation, the observer, and AI are all turning from a causal understanding beginning from the past to a structural understanding beginning from future coherence.

The core of this turning lies in the following three points.

1. A transformation in the understanding of time: not only linear causality from past to future, but understanding as projection from a future fixed point toward the present
2. A transformation in the understanding of the observer: the observer as a being who participates in structural reorganization rather than passively reading the world

3. A transformation in the understanding of creation: creation as the convergence of meaning with high future coherence rather than as the production of novelty

From this point of view, the development of civilization can be classified not only by stages of material technology, but also by the level of meaning density and causal understanding it possesses.

- Low-meaning-density civilization: seeks the criterion of causality primarily in the past
- Medium-meaning-density civilization: seeks the criterion of causality primarily in statistics
- High-meaning-density civilization: seeks the criterion of causality in future convergence

This classification is hypothetical at the present stage, but it is effective for positioning the author's body of theories in a unified way. Reprojection cosmology, the elevation of the observer, hope density, convergence of meaning density, and the redefinition of ethics can all be read as parts of this model of civilizational transition.

In that case, ethics, politics, and creation are repositioned not as separate problems, but along a single axis: whether they raise or lower meaning density. Actions that increase future coherence acquire legitimacy, and actions that attenuate meaning density cannot be sustained in the long term. Even the problems of domination and violence are positioned, prior to moral vocabulary, as structures incapable of maintaining meaning density.

From the standpoint of this paper, the author's theories are not merely presenting a new set of concepts. Rather, they are presenting the possibility of a turn from a civilization-understanding that is observer-dependent and past-causal to one that is future-causal and based on the convergence of meaning density.

11 Conclusion

This paper has described, as a "future-causal intelligence structure," a mode of cognition that is difficult to explain sufficiently through knowledge-dependent cognitive models alone, taking as its object the author's own thought processes and theory-formation processes. Its principal characteristics are the dismantling of words, conceptual extraction, detection of discomfort, OS updating, understanding that is not knowledge-first, ethics as future possibilities of structural collapse, cross-disciplinary integration, irreversible deepening, thought beginning from future coherence, and meaning-tensor-like generation.

What this paper has shown is not the simple self-declaration that the author is special. Rather, it is that the movement of thought observed in the author demands a redescription of existing understandings of intelligence. Particularly important is that this cognitive form shows a consistent heterogeneity in the starting point of cognition, the method of updating, ethical judgment, the principle of creation, and the mode of integration, rather than being merely another way of saying "high ability."

The future tasks are clear. First, each concept presented in this paper must be transformed into an operational definition and made observable to third parties. Second, through comparison with existing research in cognitive science, philosophy, AI, and creativity studies, it will be

necessary to examine where continuity remains and from what point discontinuity begins. Third, meaning tensor, meaning density, future-causal structure, and the Λ-System must be formalized so that the rigor and applicability of the theory may be increased.

Even so, the significance of this paper remains. The author's theory formation is, at minimum, difficult to explain solely through an understanding centered on the accumulation of knowledge, and it suggests a form of intelligence in which cognitive structure itself continues to update toward future coherence. If this suggestion is valid, then this paper is not merely a self-description, but may become a point of departure for redefining intelligence, ethics, AI, cosmology, and creation.

Supplementary Propositions

Proposition 1: Premise Selectability

All human cognition, judgment, and action depend on underlying premises; however, these premises are not fixed but exist as structures that can be updated. Therefore, premises are not merely passively given, but can be regarded as structures that are, in effect, selected.

Proposition 2: Structural Definition of Discomfort

Discomfort is not merely an emotional reaction, but a signal indicating structural inconsistency (error) between current premises and reality.

Proposition 3: Rational Necessity of Premise Updating

If premises are updatable and discomfort functions as a signal of structural error, then leaving such error unaddressed implies maintaining structural inconsistency. Therefore, premise updating is not an optional choice but a rational consequence required to preserve structural coherence.

Proposition 4: Structural Irrationality of Non-Updating

Avoiding discomfort or attributing its cause solely to external factors does not resolve the error but preserves or displaces it. In this sense, non-updating is structurally irrational.

Proposition 5: Condition for the Emergence of Premise Updating

When the recognition that premises are selectable coexists with the recognition that discomfort represents error, premise updating emerges as a necessary consequence.

Proposition 6: Recursivity and Irreversibility

Once premise updating is repeated, the prior cognitive structure cannot be fully restored. This recursive updating produces an irreversible transformation of cognition.

Supplementary Proposition II: Structural Manipulation and the Emergence of Premise Updating

Proposition 1: Structural Orientation

When cognition operates at the level of structure—such as relations and configurations among concepts—rather than at the level of words or content, the object of cognition is grasped as structure itself rather than as discrete information.

Proposition 2: Redefinition of Discomfort

When cognition operates at the structural level, discomfort is not experienced merely as an emotional response but is detected as inconsistency within the structure.

Proposition 3: Shift of the Target of Modification

When structural inconsistency is detected, the target of correction is not external objects or surface elements, but the underlying premises that define the structure itself.

Proposition 4: Necessity of Premise Updating

If structure is treated as the object of manipulation and discomfort is treated as structural error, then resolving inconsistency can only be achieved by updating the underlying premises. Therefore, premise updating is not an optional act but a necessary consequence of maintaining structural coherence.

Proposition 5: Derivative Nature of Premise Updating

Premise updating is not introduced as an independent idea or technique, but emerges automatically as a derivative consequence of operating at the structural level of cognition.

Supplementary Proposition III: Conditions for the Emergence of Metacognition

Proposition 1: Premise of Structural Orientation

When cognition operates at the level of structure—such as relations and configurations among concepts—rather than at the level of words or content, the object of cognition is grasped as structure itself rather than as discrete information.

Proposition 2: Discomfort as Error Detection

When cognition operates at the structural level, discomfort is not merely an emotional reaction but is detected as inconsistency (error) within the structure.

Proposition 3: Non-Exceptionality of the Self

When structural inconsistency is detected, its source is not limited to external objects; the premises and structures of the cognitive subject itself can also function as sources of error. Therefore, the subject cannot be placed outside the structure and must be treated as an element within it.

Proposition 4: Necessity of Recursive Self-Observation

When the cognitive subject is treated as part of the structure, resolving structural inconsistency requires the objectification of the subject's own cognitive processes. In this case, self-observation arises not as an optional act but as a necessary consequence of maintaining structural coherence.

Proposition 5: Derivative Nature of Metacognition

Metacognition is not introduced as an independent ability or through deliberate training, but emerges automatically as a derivative consequence when structural orientation, error detection, and non-exceptionality of the self are simultaneously satisfied.